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Altering card device attributes in response to detecting a proximity intrusion

Abstract

A card device receives a first sensor data from sensors associated with the card device and extracts a set of user features from the first sensor data, where the user features indicate physical attributes of the user. The card device receives a second sensor data from the sensors. The card device extracts a set of viewer features from the second sensor data, where the viewer features indicate physical attributes of the viewer. The card device compares at least one of the user features with a counterpart feature from among the viewer features. The card device determines that the user is not the viewer. The card device determines a distance between the viewer and the card device. The card device determines that the distance between the viewer and the card device is less than a threshold distance. In response, the card device dynamically masks the information displayed on the card device.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to information security, and more specifically to altering card device attributes in response to detecting a proximity intrusion.

BACKGROUND

(2) Cards display confidential information of the users. In some cases, the cards may be misplaced or stolen. In some cases, other people may be close to a card and be able to glance at the confidential information displayed on the card. While performing daily activities, the users are prone to scenarios where unauthorized viewers can view the confidential information displayed on the cards. It is challenging to protect confidential information displayed on the cards from unauthorized viewers. The existing systems lack capabilities to mask confidential information from unauthorized viewers in such scenarios.

SUMMARY

(3) The system described in the present disclosure is particularly integrated into practical applications of improving the information security technology by masking information displayed on the card device and/or disabling data communications for the card device in response to detecting an anomaly with respect to the card device. Examples of the anomaly may include a case where the card device is in an area that deviated from a baseline environment data profile associated with the user associated with the card device, a case where one or more unauthorized people or viewers are within a threshold proximity or distance from the card device, and the like. Certain technical advantages are provided by the embodiments of the present disclosure.

(4) In some cases, the card device may be misplaced or stolen. In some cases, some people may be close to the card device while the user is carrying the card device, and the people may be able to view the user information on the card device. The user information may include sensitive or private information associated with the user.

(5) Therefore, the disclosed system provides a solution to these and other technical problems by implementing an unconventional system and method configured to detect situations where the card device is misplaced or stolen, and cases where unauthorized people are within a threshold distance from the card device and can view the information displayed on the card device. To protect the user information from unauthorized access, the card device is configured to detect such cases and, in response, mask the user information and/or disable the data communications for the card device. For example, masking the user information may include replacing the text indicating the user information with other characters, e.g., "x". In response to disabling the data communications for the card device the information stored in the memory of the card device cannot be accessed or communicated to another device. Therefore, the user information and other information stored in the memory of the card device are protected from unauthorized access and the card device cannot be used to perform actions, e.g., communicate data to and/or from another device. In this manner, the system is integrated into a practical application of detecting anomalies with respect to the card device and, in response, protecting the information stored by the card device, and disabling the data communications for the card device.

(6) In this way, the disclosed system improves the data security technology and the security of the card device and information stored in a memory of the card device. Therefore, the disclosed system minimizes (or prevents) unauthorized access to non-public information associated with the users. This leads to securing non-public information from unauthorized access by bad actors.

(7) Altering Card Device Attributes in Response to Detecting an Anomalous Location of the Card Device

(8) In certain embodiments, a system for dynamically changing attributes of the card device comprises at least one sensor, a display field, and a processor. The at least one sensor is configured to capture sensor data, wherein the sensor data provides information about an environment around a card device. The display field is configured to display information associated with a first user. The processor is operably coupled to the at least one sensor and the display field. The processor is configured to receive a first sensor data from the at least one sensor. The processor is further configured to generate a baseline environment data profile for the first user based at least in part upon the first sensor data, wherein the baseline environment data profile indicates a geographical location where the first user typically carries the card device. The processor is further configured to receive a second sensor data from the at least one sensor. In response to receiving the second sensor data, the processor is further configured to extract baseline features from the first sensor data, wherein the baseline features provide information about the baseline environment data profile. The processor is further configured to extract test features from the second sensor data. The processor is further configured to compare at least one of the extracted baseline features with a counterpart feature from among the extracted test features. The processor is further configured to determine that the card device is in an area that deviates from the geographical location where the card device is typically carried in response to determining that the at least one of the extracted baseline features does not correspond to the counterpart feature from among the extracted test features. The processor is further configured to dynamically mask the information displayed on the display field in response to determining that the card device is in the area that deviates from the geographical location.

(9) Altering Card Device Attributes in Response to Detecting Proximity Intrusion

(10) In certain embodiments, a system for dynamically changing attributes of the card device comprises at least one sensor, a display field, and a processor. The at least one sensor is configured to capture sensor data, wherein the sensor data provides information about an environment around a card device. The display field is configured to display information associated with a first user. The processor is operably coupled to the at least one sensor and the display field. The processor is configured to receive a first sensor data from the at least one sensor, wherein the first sensor data indicates presence of the first user around the card device. The processor is further configured to extract a set of user features from the first sensor data, wherein the set of user features indicate physical attributes of the first user. The processor is further configured to receive a second sensor data from the at least one sensor, wherein the second sensor data indicates presence of a viewer around the card device. The processor is further configured to extract a set of viewer features from the second sensor data, wherein the set of viewer features indicate physical attributes of the viewer. The processor is further configured to compare at least one of the user features with a counterpart feature from among the set of viewer features. The processor is further configured to determine that the more than a threshold percentage of the set of user features does not correspond to counterpart features from among the set of viewer features. The processor is further configured to determine that the viewer is not the first user in response to determining that more than the threshold percentage of the set of baseline features does not correspond to the counterpart features from among the set of viewer features. The processor is further configured to determine a distance between the viewer and the card device. The processor is further configured to compare the determined distance with a threshold distance. The processor is further configured to determine that the determined distance is less than the threshold distance. The processor is further configured to dynamically mask the information displayed on the display field in response to determining that the determined distance is less than the threshold distance and that the viewer is not the first user.

(11) Certain embodiments of this disclosure may include some, all, or none of these advantages.

These advantages and other features will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.

(2) FIG. 1 illustrates an embodiment of a system configured to alter card device attributes in response to an anomaly;

(3) FIG. 2 illustrates an example operational flow of the system of FIG. 1 for altering card device attributes in response to an anomalous location of the card device;

(4) FIG. 3 illustrates an example flowchart of a method for altering card device attributes in response to an anomalous location of the card device;

(5) FIG. 4 illustrates an example operational flow of the system of FIG. 1 for altering card device attributes in response to a proximity intrusion; and

(6) FIG. 5 illustrates an example flowchart of a method for altering card device attributes in response to a proximity intrusion.

DETAILED DESCRIPTION

(7) As described above, previous technologies fail to provide efficient and reliable solutions to detect anomalies with respect to a card device. Embodiments of the present disclosure and its advantages may be understood by referring to FIGS. 1 through 5. FIGS. 1 through 5 are used to describe systems and methods to alter card device attributes in response to an anomaly according to certain embodiments.

(8) System Overview

(9) FIG. 1 illustrates an embodiment of a system **100** that is generally configured to alter attributes of a card device **120** in response to detecting an anomaly with respect to the card device **120**, where the anomaly may include a case where an unauthorized person is within a threshold distance from the card device **120** and can identity the information displayed on the card device **120** and a case where the card device **120** is in an area/location that deviates from the known locations where the card device **120** is typically carried. In certain embodiments, the system **100** comprises the card device **120** communicatively couples to a backend server **150** and a blockchain network **180** via a network **110**. Network **110** enables the communications of the components of the system **100**. The card device **120** includes a processor **122** in signal communication with a memory **130**. The memory **130** stores software instructions **132** that when executed by the processor **122** cause the processor **122** to perform one or more operations described herein. The backend server **150** includes a processor **152** in signal communication with a memory **156**. Memory **156** stores software instructions **158** that when executed by the processor **152** cause the processor **152** to perform one or more operations described herein. In other embodiments, system **100** may not have all of the components listed and/or may have other elements instead of, or in addition to, those listed above.

(10) In general, the system **100** improves information security technology by masking information **138** displayed on the card device **120** and/or disabling data communications for the card device **120** in response to detecting an anomaly with respect to the card device **120**. Examples of the anomaly may include a case where the card device **120** is in an area that deviated from a baseline environment data profile **144** associated with the user **102** associated with the card device **120**, a case where one or more unauthorized people or viewers are within a threshold proximity or distance **140** from the card device **120**, and the like. The operation by which the card device **120**

may alter the attributes of the card device **120** in response to detecting an anomalous location of the card device **120** is described in FIGS. 2-3. The operation by which the card device **120** may alter the attributes of the card device **120** in response to detecting a proximity intrusion of a viewer is described in FIGS. 4-5.

(11) In some cases, the card device **120** may be misplaced or stolen. In some cases, some people may be close to the card device **120** while the user **102** is carrying the card device **120**, and the people may be able to view the user information **138** on the card device **120**. The user information **138** may include sensitive or private information associated with the user **102**. To protect the user information **138** from unauthorized access, the card device **120** is configured to detect such cases and, in response, mask the user information **138** and/or disable the data communications for the card device **120**. For example, masking the user information **138** may include replacing the text indicating the user information **138** with other characters, e.g., “x”. In response to disabling the data communications for the card device **120**, the information stored in the memory **130** of the card device **120** cannot be accessed or communicated to another device. Therefore, the user information **138** and other information stored in the memory **130** of the card device **120** are protected from unauthorized access and the card device **120** cannot be used to perform actions, e.g., communicate data to and/or from another device. In this manner, the system **100** is integrated into a practical application of detecting anomalies with respect to the card device **120** and, in response, protecting the information stored by the card device **120**, and disabling the data communications for the card device **120**. Therefore, system **100** provides a solution to technical problems of card devices **120** being misplaced or stolen, and unauthorized people being within a threshold distance **140** from the card device **120**.

(12) System Components

(13) Network

(14) Network **110** may be any suitable type of wireless and/or wired network. The network **110** may be connected to the Internet or public network. The network **110** may include all or a portion of an Intranet, a peer-to-peer network, a switched telephone network, a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), a personal area network (PAN), a wireless PAN (WPAN), an overlay network, a software-defined network (SDN), a virtual private network (VPN), a mobile telephone network (e.g., cellular networks, such as 4G or 5G), a plain old telephone (POT) network, a wireless data network (e.g., WiFi, WiGig, WiMAX, etc.), a long-term evolution (LTE) network, a universal mobile telecommunications system (UMTS) network, a peer-to-peer (P2P) network, a Bluetooth network, a near-field communication (NFC) network, and/or any other suitable network. The network **110** may be configured to support any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(15) Card Device

(16) A card device **120** is a portable card device that is generally configured to process data and interact with users. Examples of the card device **120** include a personal card, etc. In the illustrated embodiment, the card device **120** includes a processor **122** in signal communication with a network interface **124**, sensors **126**, a display field **128**, and a memory **130**. In certain embodiments, the card device **120** may be configured as shown or other configurations.

(17) Processor **122** comprises one or more processors. The processor **122** is any electronic circuitry, including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate arrays (FPGAs), application-specific integrated circuits (ASICs), or digital signal processors (DSPs). For example, one or more processors may be implemented in cloud devices, servers, virtual machines, and the like. The processor **122** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable number and combination of the preceding. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **122** may be 8-bit, 16-bit, 32-bit, 64-bit, or of any other suitable architecture. The processor **122** may

include an arithmetic logic unit (ALU) for performing arithmetic and logic operations. The processor **122** may register the supply operands to the ALU and store the results of ALU operations. The processor **122** may further include a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers, and other components. The one or more processors are configured to implement various software instructions. For example, the one or more processors are configured to execute instructions (e.g., software instructions **132**) to perform the operations of the card device **120** described herein. In this way, processor **122** may be a special-purpose computer designed to implement the functions disclosed herein. In an embodiment, the processor **122** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The processor **122** is configured to operate as described in FIGS. **1-5**. For example, the processor **122** may be configured to perform one or more operations of the operational flow **200** as described in FIG. **2**, one or more operations of method **300** as described in FIG. **3**, perform one or more operations of the operational flow **400** as described in FIG. **4**, and one or more operations of method **500** as described in FIG. **4**.

(18) Network interface **124** is configured to enable wired and/or wireless communications. The network interface **124** may be configured to communicate data between the card device **120** and other devices, systems, or domains. For example, the network interface **124** may comprise an NFC interface, a Bluetooth interface, a Zigbee interface, a Z-wave interface, a radio-frequency identification (RFID) interface, a WIFI interface, a local area network (LAN) interface, a wide area network (WAN) interface, a metropolitan area network (MAN) interface, a personal area network (PAN) interface, a wireless PAN (WPAN) interface, a modem, a switch, and/or a router. The processor **122** may be configured to send and receive data using the network interface **124**. The network interface **124** may be configured to use any suitable type of communication protocol.

(19) Sensors **126** may include any type of sensor, including one or more cameras, sound sensors, light sensors, humidity sensors, biometric sensors, pressure sensors, motion sensors, a global positioning system (GPS) sensor, and one or more microelectromechanical system (MEMS) sensors, among others. Each sensor **126** may be configured to detect respective sensor data **134**. For example, the cameras may be configured to capture images and/or videos, the sound sensors may be configured to capture audio signals, the light sensors may be configured to detect light brightness levels, the humidity sensors may be configured to detect humidity levels, the biometric sensors may be configured to detect biometric features (e.g., fingerprint, etc.), the pressure sensors may be configured to detect air pressure, e.g., with respect to the sea level, the motion sensors may be configured to detect movements of the objects, the GPS sensor may be configured to detect the location coordinate of the card device **120**, and the MEMS sensors may be configured to detect any attribute with respect to their surroundings.

(20) The sensor data **134** may include an image feed, a video feed, an audio feed, a light data feed (that indicates the light brightness levels), a humidity data feed (that indicates the humidity levels), a biometric data feed (that indicates the biometric features), a pressure data feed (that indicates the pressure levels), a motion data feed (that indicates movements of objects), a location data feed, and any other type of data feed. The sensor data **134** may provide information about the environment surrounding the card device **120** (e.g., within a detection range or field of view of the sensors **126**). For example, the sensor data **134** may provide information about objects around the card device **120**, such as shapes, colors, locations, etc.

(21) Display field **128** may be a display portion of the card device **120** and is generally configured to display user information **138**. The display field **128** may include a liquid crystal display (LCD), a light emitting diode (LED) display, and the like. For example, the display field **128** may display a name, a number, an address, etc. associated with the user **102**.

(22) Memory **130** may be volatile or non-volatile and may comprise read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM). The memory **130** may

include one or more of a local database, a cloud database, a network-attached storage (NAS), etc. The memory **130** comprises one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **130** may store any of the information described in FIGS. **1-5** along with any other data, instructions, logic, rules, or code operable to implement the function(s) described herein when executed by processor **122**. For example, the memory **130** may store software instructions **132**, user profile **136**, user information **138**, test features **212**, baseline features **210**, user features **404**, viewer features **408**, sensor data **134a-d**, baseline environment data profile **144**, interaction session data **146**, machine learning algorithm **142**, threshold distance **140**, pose estimation data **220**, **224**, messages **214**, **216**, and/or any other data or instructions. The software instructions **132** may comprise any suitable set of instructions, logic, rules, or code operable to execute the processor **122** and perform the functions described herein, such as some or all of those described in FIGS. **1-5**.

(23) Machine learning algorithm **142** may be implemented by the processor **122** executing software instructions **132**, and is generally configured to generate the baseline environment data profile **144** and other environment data based on the sensor data **134**, determine an identity of the user **102** based on biometric data associated with the user **102**, such as image feed, audio feed, pose estimation, etc., determine whether the card device **120** is misplaced or stolen, among one or more other operations described herein.

(24) In certain embodiments, the machine learning algorithm **142** may include a support vector machine, neural network, random forest, k-means clustering, Tree-based algorithm, Random Forest algorithm, etc. In certain embodiments, the machine learning algorithm **142** may include image processing, object detection, object recognition, video processing algorithms, biometric feature processing, and the like.

(25) The machine learning algorithm **142** may be implemented by supervised, semi-supervised, or unsupervised training algorithms. In a training stage, the machine learning algorithm **142** may be given a training dataset comprising a set of sample data (e.g., an image, an audio, a video, a biometric sample, etc.) each labeled with a respective label data. For example, the machine learning algorithm **142** may be given a set of images of the user **102** and a label data indicating the identity of the user **102**. The machine learning algorithm **142** may extract a set of features from the sample data and associated the features with the label data. In this process, the machine learning algorithm **142** may learn the correlation between each sample data and the respective label data. In the testing stage, the machine learning algorithm **142** may be given a test data (e.g., an image, an audio, a video, a biometric sample, etc.) that is not labeled with a label data, and is asked to predict the label data associated with the test data. The machine learning algorithm **142** may extract a set of features from the test data and compare the extracted features with the training dataset. The machine learning algorithm **142** may determine to which sample data the test data corresponds to based on the comparison. In response to finding a matching or corresponding features from the training dataset, the machine learning algorithm **142** may determine that the test data is associated with a particular label data associated with the sample data with the matching or corresponding features. In a refining and backpropagation, the machine learning algorithm **142** may update the weight and bias values of the neural networks to increase the accuracy of the prediction of the machine learning algorithm **142**.

(26) The baseline environment data profile **144** comprises data about one or more geographical locations where the user **102** carries the card device **120**. For example, the baseline environment data profile **144** may include sensor data **134** associated with the one or more geographical locations where the user **102** carries the card device **120**. For example, the baseline environment data profile **144** may include an image feed, an audio feed, a humidity data, pressure data (e.g., air pressure with respect to the seal level), light data, GPS location coordinate data, and motion data captured by the camera, a sound sensor, a humidity sensor, a pressure sensor, a light sensor, a GPS

sensor, a motion sensor associated with the one or more geographical location, respectively.

(27) The data about the user **102** may be determined by the machine learning algorithm **142** from the sensor data **134**, including an image of the user **102**, an audio signal associated with the user **102**, etc. The user profile **136** may include the user information **138** associated with the user **102**. The user information **138** may include information associated with the user **102** and displayed on the display field **128**. For example, the user information **138** may include a name, an address, a number, a security code a calendar date, an expiration date, etc. associated with the user **102**. The user features **404** may comprise data about the user **102**, such as facial features, audio features, biometric features, pose estimation, etc. The viewer features **408** may comprise data about the viewer (**402** in FIG. 4), such as facial features, audio features, biometric features, pose estimation, etc. The viewer features **408** and the user features **404** may be determined by the machine learning algorithm **142**. These operations are described in greater detail in FIGS. 2-5.

(28) Backend Server

(29) Backend server **150** generally includes a hardware computer system configured to store the user profiles **136** and corresponding baseline environment data profile **144**. In certain embodiments, the backend server **150** may be implemented by a cluster of computing devices, such as virtual machines. For example, the backend server **150** may be implemented by a plurality of computing devices using distributed computing and/or cloud computing systems in a network. In certain embodiments, the backend server **150** may be configured to provide services and resources to other components and devices.

(30) The backend server **150** comprises a processor **152** operably coupled with a network interface **154** and a memory **156**. Processor **152** comprises one or more processors. The processor **152** is any electronic circuitry, including, but not limited to, state machines, one or more CPU chips, logic units, cores (e.g., a multi-core processor), FPGAs, ASICs, or DSPs. For example, one or more processors may be implemented in cloud devices, servers, virtual machines, and the like. The processor **152** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable number and combination of the preceding. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **152** may be 8-bit, 16-bit, 32-bit, 64-bit, or of any other suitable architecture. The processor **152** may include an ALU for performing arithmetic and logic operations. The processor **152** may register the supply operands to the ALU and store the results of ALU operations. The processor **152** may further include a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers, and other components. The one or more processors are configured to implement various software instructions. For example, the one or more processors are configured to execute instructions (e.g., software instructions **158**) to perform the operations of the backend server **150** described herein. In this way, processor **152** may be a special-purpose computer designed to implement the functions disclosed herein. In an embodiment, the processor **152** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The processor **152** is configured to operate as described in FIGS. 1-5. For example, the processor **152** may be configured to perform one or more operations of the operational flow **200** as described in FIG. 2, one or more operations of method **300** as described in FIG. 3, perform one or more operations of the operational flow **400** as described in FIG. 4, and one or more operations of method **500** as described in FIG. 4.

(31) Network interface **154** is configured to enable wired and/or wireless communications. The network interface **154** may be configured to communicate data between the backend server **150** and other devices, systems, or domains. For example, the network interface **154** may comprise an NFC interface, a Bluetooth interface, a Zigbee interface, a Z-wave interface, an RFID interface, a WIFI interface, a LAN interface, a WAN interface, a MAN interface, a PAN interface, a WPAN interface, a modem, a switch, and/or a router. The processor **152** may be configured to send and receive data using the network interface **154**. The network interface **154** may be configured to use any suitable

type of communication protocol.

(32) The memory **156** may be volatile or non-volatile and may comprise ROM, RAM, TCAM, DRAM, and SRAM. The memory **156** may include one or more of a local database, a cloud database, a NAS, etc. The memory **156** comprises one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **156** may store any of the information described in FIGS. **1-5** along with any other data, instructions, logic, rules, or code operable to implement the function(s) described herein when executed by processor **152**. For example, the memory **156** may store software instructions **158**, user profiles **136**, baseline environment data profiles **144**, and/or any other data or instructions. The software instructions **158** may comprise any suitable set of instructions, logic, rules, or code operable to execute the processor **152** and perform the functions described herein, such as some or all of those described in FIGS. **1-5**.

(33) Blockchain Network

(34) Blockchain network **180** is a peer-to-peer network of network nodes, and is generally configured to distribute interaction session data **146** (and any other data/information) among the network nodes **182a-n**. In certain embodiments, the blockchain network **180** is a distributed database in a network of network nodes **182a-n**. In certain embodiments, blockchain network **180** may be a public blockchain network. In certain embodiments, blockchain network **180** may be a private blockchain network. For example, membership in blockchain network **180** may be limited to nodes registered as belonging to and/or affiliated with the organization to which the backend server **150** and/or the card device **120** belongs. In certain embodiments, the backend server **150** may be a member of blockchain network **180** (e.g., as nodes among nodes **182a-n** in blockchain network **180**).

(35) The blockchain network **180** may comprise any number of network nodes **182a-n** to form a distributed network that maintains a blockchain **186**. Each network node **182** may comprise a computing device, a virtual machine, a server, a workstation, and/or the like. Each network node **182a** through **192n** of blockchain network **180** stores a blockchain database **184** that is configured to store a copy of a blockchain **186**. Each network node **182a** through **192n** may be an instance of a network node **182**. The network node **182** may include a hardware processor, memory, and/or circuitry (not explicitly shown) configured to perform any of the functions or actions of the network node **182** described herein. For example, a software application designed using software code may be stored in the memory and executed by the processor to perform the functions of the network node **182**. The network node **182** is configured to communicate with other devices and components of the system **100** via the network **110**.

(36) In certain embodiments, the backend server **150** may access the blockchain network **180** via the network **110** to distribute information in the blockchain network **180**, including the interaction session data **146**. The blockchain network **180** may be configured to distribute interaction session data **146** among the nodes. The backend server **150** may use an application programming interface (API) service via the network **110** to access the blockchain network **180**.

(37) The blockchain network **180** is configured to establish consensus among the network nodes **182a-n** about the present state of the blockchain database **184**. For example, each network node **182a-n** comprises a processor in signal communication with a memory storing software instructions that when executed by the processor, cause the network nodes **182a-n** to implement a consensus protocol procedure through which all the network nodes **182a-n** of the blockchain network **180** reach a common agreement about the present state of the blockchain database **184**. In this way, each network node **182a-n** achieves reliability in the blockchain network **180** and establishes trust between the network nodes **182a-n** in a distributed computing environment. Essentially, the consensus protocol makes sure that every new block that is added to the blockchain **186** is the one and only version of the truth that is agreed upon by all the block in the blockchain

186. Blockchain **186** links together blocks of data, which store identifiable units called blockchain data entries. The blockchain data entry may be interchangeably referred to herein as a blockchain data entry. The blockchain data entries stored in the blockchain **186**, may include information, files, and/or any other suitable type of data. For example, blockchain data entries may include interaction session data **146** received from the backend server **150**. Each interaction session data **146** may be associated with a different interaction session. The interaction session data **146** may include information about a sending entity (e.g., the card device **120** or another device), information about a receiving entity (e.g., the card device **120** or another device), authentication token, data transferred (e.g., software and/or hardware services), a timestamp of each data transfer, and/or any other information, for example. With every new data transfer (e.g., every new interaction session), a new block of data may be generated and added to the blockchain **186**.

(38) Operational Flow for Altering Attributes of the Card Device in Response to Detecting an Anomalous Location of the Card Device

(39) FIG. 3 illustrates an example operational flow **200** of the system **100** of FIG. 1 for altering attributes of the card device **120** in response to detecting anomalous location of the card device **120** according to one or more embodiments. In operation, the operational flow **200** may begin when the card device **120** generates the baseline environment data profile **144**.

(40) Generating the Baseline Environment Data Profile

(41) For example, assume that the user **102** registers at an organization and obtains the card device **120**. Upon registering and obtaining the card device **120**, the user **102** may authenticate themselves to the card device **120**. For example, the user **102** may setup credentials with the card device **120**, e.g., by connecting the card device **120** to a computer system and setting up the credentials via an interface application associated with the card device **120**. In another example, the user **102** may setup the credentials with the card device **120** via the sensors **126**, e.g., by providing fingerprints. In another example, the card device **120** may be paired with the user **102** by capturing an image of the user **102** (by a camera sensor **126**), extracting facial features of the user **102** (e.g., via the machine learning algorithm **142**), and associating the facial features of the user **102** to the card device **120**. In other examples, any attribute associated with the user **102** may be used to pair the user **102** to the card device **120**, such as an audio sample, pose estimation, retinal feature, etc.

(42) The card device **120** is to be trained to understand the geographical locations where the user **102** typically carries the card device **120**. For example, the geographical locations where the user **102** typically carries the card device **120** may include the house, office, vehicle, a pathway where the user **102** walks in the streets, among others. To this end, in the training phase, the sensors **126** may detect the presence of objects **202** around the card device **120**. The card device **120** may receive first sensor data **134a** from the sensors **126**, where the first sensor data **134a** may indicate presence of first set of objects **202** around the card device **120** within a detection range of the sensors **126**. This process may be performed at one or more locations where the user **102** wants the card device **120** to be trained to learn the environment, such as inside the house of the user **102**, inside the vehicle of the user **102**, inside the office of the user **102**, etc. For example, the sensors **126** may detect the presence of the user **102**, objects inside the house of the user **102**, objects inside the vehicle of the user **102**, objects inside the office of the user **102**, etc.

(43) The first sensor data **134a** is fed to the machine learning algorithm **142**. The machine learning algorithm **142** may process the first sensor data **134a** and generate the baseline environment data profile **144** based on the sensor data **134a**. The baseline environment data profile **144** may indicate geographical location(s) where the user **102** typically carries the card device **120**, such as the house, vehicle, particular pathways where the user **102** walks, office, etc.

(44) In certain embodiments, the machine learning algorithm **142** may generate the baseline environment data profile **144** in response to extracting a set of baseline features **210** from the sensor data **134a** and associating that baseline features **210** to the user **102**. For example, the baseline features **210** may provide information about the baseline environment data profile **144**

(e.g., the geographical location(s) where the user **102** typically carries the card device **120**). In other words, the baseline features **210** may provide information about objects **202** detected by the sensors **126** when the card device **120** is in geographical location(s) where the user **102** typically carries the card device **120**. The baseline features **210** may indicate physical attributes of the objects **202**, such as shapes, colors, and locations. In response, the card device **120** may be trained to understand that the user **102** typically carries the card device **120** in the geographical location(s) indicated in the baseline environment data profile **144**. The card device **120** may perform similar operations for each location that the user **102** carries the card device **120**, e.g., the house, vehicle, office, pathways where the user **102** walks, etc. For example, when the user **102** carries the card device **120** to their house, the card device **120** may prompt the user **102** to authenticate themselves, e.g., by providing a biometric sample, via an image of the user, etc. by providing a user credential, e.g., username password, a security code sent to a mobile device associated with the user **102**, among others.

(45) In response to the user **102** being authenticated, the sensors **126** capture sensor data **134a** and communicate the sensor data **134a** to the processor of the card device **120**. The card device **120** processes the sensor data **134a**, detects the objects **202** inside the house of the user **102**, associate the house of the user **102** to the user **102**, and includes the house of the user **102** in the baseline environment data profile **144** for the user **102**. The card device **120** may perform similar operations for other locations where the user **102** carries the card device **120** and authenticate themselves to the card device **120**, similar to that described above. In this manner, the card device **120** is trained to identify and understand the environments or locations where the user **102** typically carries the card device **120** and include those in the baseline environment data profile **144**.

(46) In certain embodiments, if a new environment is detected, the card device **120** may prompt the user **102** to authenticate themselves, e.g., by displaying an authentication request on the display field. If the user **102** is authenticated by the card device **120**, the card device **120** may perform similar operations to understand and identify the new environment based on the objects in the new environment and add the new environment to the baseline environment data profile **144**.

(47) In certain embodiments, the card device **120** may disregard (and delete) a geographical location from the baseline environment data profile **144** if it is determined that the user **102** has not carried the card device **120** in the geographical location for more than a threshold period, e.g., more than five months, three months, etc. For example, the card device **120** may keep the most recent geographical locations that the user **102** carried the card device **120** within past five months, three months, etc. in the baseline environment data profile **144**. In this manner, whenever the card device **120** is carried to one of the known geographical locations indicated in the baseline environment data profile **144**, the card device **120** may recognize that it is one of the known locations, and not an anomalous (e.g., unfamiliar) location.

(48) Evaluating a New Environment Against the Baseline Environment Data Profile

(49) The card device **120** may determine if a new environment (that the card device **120** is located) corresponds to one of the known geographical locations included in the baseline environment data profile **144**. For example, assume that the card device **120** is carried to a new location. The sensors **126** may capture second sensor data **134b**, where the second sensor data **134b** may indicate presence of a second set of objects **204** around the card device **120**. For example, the second set of objects **204** may include objects in the new location. The sensor data **134b** may indicate locations, shapes, and colors of the second set of objects **204**.

(50) The card device **120** may receive the second sensor data **134b** from the sensors **126**. In response, the card device **120** may access the baseline features **210** associated with the baseline environment data profile **144**. In certain embodiments, the card device **120** may extract the baseline features **210** from the sensor data **134a** in response to receiving the sensor data **134b**. The card device **120** may also feed the sensor data **134b** to the machine learning algorithm **142** for processing, similar to that described above. The card device **120** (e.g., via the machine learning

algorithm **142**) may extract the test features **212** from the sensor data **134b**. The test features **212** may indicate physical attributes of the objects **204**, such as shapes, colors, and locations.

(51) The card device **120** may compare the baseline features **210** with the test features **212** to determine whether the new location corresponds to any of the known locations in the baseline environment data profile **144**. In certain embodiments, the baseline features **210** may be represented by a baseline feature vector **226** comprising numerical values, and the test features **212** may be represented by test feature vector **228** comprising numerical values.

(52) In certain embodiments, comparing the extracted baseline features **210** with the counterpart feature from among the extracted test features **212** may be in response to comparing at least one (or at least more than a threshold percentage) of the numerical values of the baseline feature vector **226** with a counterpart numerical value of the test feature vector **228**.

(53) In certain embodiments, determining that the card device **120** is in the area that deviates from the geographical location where the card device **120** is typically carried (as indicated in the baseline environment data profile **144**) may be in response to determining that the at least one (or at least more than a threshold percentage, e.g., more than 50%, 60%, etc.) of the numerical values of the baseline feature vector **226** does not correspond to the counterpart numerical value in the test feature vector **228**.

(54) In certain embodiments, the card device **120** may compare each of the baseline features **210** with a counterpart test feature from among the test features **212**. The card device **120** may determine that the card device **120** is in an area that deviates from the geographical locations where the user **102** typically carries the card device **120** (as indicated in the baseline environment data profile **144**) if it is determined that at least a threshold percentage of the baseline features **210** (e.g., 50%, 80%, etc.) does not correspond to the counterpart test feature from among the set of test features **212**.

(55) In certain embodiments, card device **120** may determine that the card device **120** is in an area that deviates from the geographical locations where the user **102** typically carries the card device **120** (as indicated in the baseline environment data profile **144**) if it is determined that at least one of the baseline features **210** does not correspond to the counterpart test feature from among the set of test features **212**. For example, if it is determined that a GPS location of the card device **120** is outside of the boundary of the known locations in the baseline environment data profile **144**, the card device **120** may determine that the card device **120** is in an area that deviates from the baseline environment data profile **144**.

(56) In certain embodiments, the card device **120** may determine a Euclidean distance between the baseline feature vector **226** and the test feature vector **228**. The card device **120** may determine whether the Euclidean distance is less than a threshold distance (e.g., less than 0.01, 0.1, etc.). In certain embodiments, card device **120** may determine that the card device **120** is in an area that deviates from the geographical locations where the user **102** typically carries the card device **120** (as indicated in the baseline environment data profile **144**) if it is determined that the determined Euclidean distance is more than the threshold distance.

(57) In certain embodiments, determining that the card device **120** is in an area that deviates from the geographical locations where the user **102** typically carries the card device **120** may correspond to determining that the card device **120** is outside of any of the known locations indicated in the baseline environment data profile **144**.

(58) In certain embodiments, determining that the card device **120** is in an area that deviates from the geographical locations where the user **102** typically carries the card device **120** may correspond to determining that the card device **120** is outside of a geofence area representing the known location indicated in the baseline environment data profile **144**.

(59) In response to determining that the card device **120** is in the area that deviates from the known geographical locations indicated in the baseline environment data profile **144**, the card device **120** may dynamically mask the user information **138**. For example, the card device **120** may replace the

text in the user information **138** with “x” or other characters. In another example, the user information **138** may be redacted by a block.

(60) In certain embodiments, the card device **120** may detect that a message **216** is received at the card device **120** after it is determined that the card device **120** is in the area that deviates from the known geographical locations indicated in the baseline environment data profile **144**. The message **216** may include a security token, or any other information. In response, the card device **120** may dynamically mask the received message **216**.

(61) In certain embodiments, in response to determining that the card device **120** is in the area that deviates from the baseline environment data profile **144**, the card device **120** may disable the card device **120** from being used for data communication with another device. If it is determined that the card device **120** is in the area that corresponds to a known geographical location indicated in the baseline environment data profile **144**, the card device **120** may allow the data communications with other devices upon authenticating the user **102**, and not mask or unmask the user information **138**. In certain embodiments, the user information **138** may be masked or not known and the data communications may be disabled by default, and in response to detecting that the user **102** wants to use the card device **120**, the user information **138** may be displayed and the data communications may be enabled.

(62) Adding a New Environment to the Baseline Environment Data Profile

(63) The card device **120** may add a new location to the known locations in the baseline environment data profile **144**. For example, assume that the card device **120** is carried to a new location. The sensors **126** detect a third set of objects **204** and capture sensor data **134c**. The sensors **126** communicate the sensor data **134c** to the processor of the card device **120**. The card device **120** receives the third sensor data **134c** from the sensors **126**. The sensor data **134c** indicate presence of the third set of objects **204**. In response, the card device **120** may feed the sensor data **134c** to the machine learning algorithm **142**. The card device **120** (e.g., via the machine learning algorithm **142**) may extract the second test features **212** from the third sensor data **134c**, where the second test features indicate physical attributes of the objects **204**.

(64) The card device **120** may compare each of the baseline features **210** with the counterpart test feature from the test features **212**. The card device **120** may determine whether the card device **120** is in the area that deviates from the known locations in the baseline environment data profile **144**. The card device **120** may determine that the card device **120** is in the area that deviates from the known locations in the baseline environment data profile **144** if at least one (or more than a threshold percentage) of the baseline features **210** does not correspond to the counterpart test feature **212**.

(65) In certain embodiments, the card device **120** may determine a Euclidean distance between the baseline feature vector **226** and the test feature vector **228**, and determine that the card device **120** is in the area that deviates from the known locations in the baseline environment data profile **144** if the Euclidean distance is less than a threshold distance, similar to that described above.

(66) The card device **120** may display a message **214** that requests an identity of the user **102** on the display field **128**. The user **102** may identify and authenticate themselves, e.g., by providing user credentials and/or biometric sample, similar to that described above. The card device **120** may receive the user input comprising the credential associated with the user **102**. The card device **120** may compare the received credential with previously provide and stored credential. The card device **120** may determine if the received credential corresponds to the expected credential.

(67) In response to determining that received credential corresponds to the expected credential, the card device **120** may generate a new environment data profile for the user **102**. The new environment data profile may indicate objects **204** in the new location. The card device **120** may add the new environment data profile to the baseline environment data profile **144**.

(68) Identifying the User Based on the Pose Estimation

(69) In certain embodiments, the card device **120** may determine whether the user **102** or another

person is carrying the card device **120** or accessing the card device **120**. In this process, the card device **120** may determine a first pose estimation data **220** associated with the user **102** based on processing the sensor data **134a**, where the sensor data **134a** may include an image of the user **102**, a motion data associated with the movement of the user **102**, etc. If the card device **120** is within a threshold distance **140** of another person **222**, the card device **120** may receive a fourth sensor data **134d** from the sensors **126**, where the sensor data **134d** may comprise information about the other person **222**. For example, the sensor data **134d** may comprise an image feed that provides information about a pose of the person **222**.

(70) The card device **120** may feed the sensor data **134a** and **d** to the machine learning algorithm **142**. The card device **120** (e.g., via the machine learning algorithm **142**) may determine the pose estimation data **220** from the sensor data **134a**, and determine the pose estimation data **224** associated with the person **222** from the sensor data **134d**. The card device **120** may compare the pose estimation data **220** with the pose estimation data **224**.

(71) In certain embodiments, the first pose estimation data **220** may be represented by a first pose estimation vector **230** comprising numerical values, and the second pose estimation data **224** may be represented by a second pose estimation vector **232** comprising numerical values. The card device **120** may determine that the first pose estimation data **220** does not correspond to the second pose estimation data **224**. In certain embodiments, determining that the first pose estimation data **220** does not correspond to the second pose estimation data **224** may be in response to comparing the first pose estimation vector **230** with the second pose estimation vector **232** and determining that more than a threshold number of numerical values in the first pose estimation vector **230** does not correspond to the counterpart numerical values of the second pose estimation vector **232**. In response, the card device **120** may dynamically mask the user information **138** on the display field **128** and/or disable the data communications for the card device **120**. Otherwise, the card device **120** may allow the data communications for the card device **120**.

(72) Method for Altering Card Device Attributes in Response to an Anomalous Location

(73) FIG. 3 illustrates an example flowchart of a method **300** for altering card device **120** attributes in response to an anomalous location according to certain embodiments of the present disclosure. Modifications, additions, or omissions may be made to method **300**. Method **300** may include more, fewer, or other operations. For example, operations may be performed in parallel or in any suitable order. While at times discussed as the system **100**, card device **120**, or components of any of thereof performing operations, any suitable system or components of the system may perform one or more operations of the method **300**. For example, one or more operations of method **300** may be implemented, at least in part, in the form of software instructions **132**, **158** of FIG. 1, stored on non-transitory, tangible, machine-readable media (e.g., memory **130**, **156** of FIG. 1) that when run by one or more processors (e.g., processor **122**, **152** of FIG. 1) may cause the one or more processors to perform operations **302-322**.

(74) At operation **302**, the card device **120** receives a first sensor data **134a** from at least one sensor **126**, where the first sensor data **134a** indicates presence of the first set of objects **202** around the card device **120**. For example, the sensors **126** may capture sensor data **134a** continuously, periodically, or on demand, or in response to a trigger event when the sensors **126** detect an object within their detection range.

(75) At operation **304**, the card device **120** generates the baseline environment data profile **144** based on the first sensor data **134a**. For example, the card device **120** may feed the sensor data **134a** to the machine learning algorithm **142** to generate the baseline environment data profile **144**, similar to that described in FIG. 2.

(76) At operation **306**, the card device **120** receives a second sensor data **134b** from the at least one sensor **126**, where the second sensor data **134b** indicates presence of a second set of objects **204** around the card device **120**. For example, the sensors **126** may capture the sensor data **134b** continuously, periodically, on demand, or in response to a trigger event when the sensors **126** detect

an object within their detection range.

(77) At operation **308**, the card device **120** extracts baseline features **210** from the first sensor data **134a**, e.g., via the machine learning algorithm **142**, similar to that described in FIG. 2. At operation **310**, the card device **120** extracts test features **212** from the sensor data **134b**, via the machine learning algorithm **142**, similar to that described in FIG. 2.

(78) At operation **312**, the card device **120** compares the baseline features **210** with the test features **212**. For example, the card device **120** may compare each baseline feature **210** with a counterpart test feature **212**. In another example, the card device **120** may compare the baseline feature vector with the test feature vector, similar to that described in FIG. 2.

(79) At operation **314**, the card device **120** determines whether more than a threshold percentage of the baseline features **210** correspond to the counterpart test features **212**. If it is determined that more than a threshold percentage (e.g., more than 80%, 90%, etc.) of the baseline features **210** correspond to the counterpart test features **212**, method **300** proceeds to the operation **316**.

Otherwise, method **300** proceeds to operation **318**.

(80) At operation **316**, the card device **120** may allow data communications for the card device **120**. For example, the card device **120** may allow the card device **120** to communicate data with other devices. The card device **120** may also not mask or unmask the user information **138** displayed on the card device **120**. In response, if the user **102** wants to use the card device **120** to interact with another device, the card device **120** may communicate data (e.g., user information **138**) to the other device and generate the interaction session data **146**. The card device **120** may communicate the interaction session data **146** to the backend server **150** and/or to the blockchain network **180** to be distributed among the notes **182a-n**.

(81) At operation **318**, the card device **120** determines that the card device **120** is in the area that deviates from the geographical locations that are indicated in the baseline environment data profile **144**.

(82) At operation **320**, the card device **120** dynamically masks the user information **138** displayed on the card device **120**. For example, the card device **120** may replace text of the user information **138** to other characters, e.g., "x". In another example, the card device **120** may encrypt the user information **138** by an encryption function. The user **102** may be provided the decryption key to decrypt the encrypted user information. At operation **322**, the card device **120** disables the data communication for the card device **120**. In response, the card device **120** may not be able to interact with other devices.

(83) Altering Card Device Attributes in Response to Proximity Intrusion

(84) FIG. 4 illustrates an example operational flow **400** of system **100** of FIG. 1 for altering attributes of the card device **120** in response to proximity intrusion by other (unauthorized) viewers **402** around the card device **120**. In operation, the card device **120** may establish a user profile for the user **102**. The user profile may include user features **404**. In this process, the card device **120** may receive first sensor data **134a** from the sensors **126**. The first sensor data **134a** may indicate presence of the first user **102** around the card device **120**. For example, the first sensor data **134a** may include an image of the user **102**, an audio sample of the user **102**, etc.

(85) The card device **120** may feed the sensor data **134a** to the machine learning algorithm **142**. The card device **120** (e.g., via the machine learning algorithm **142**) may extract a set of user features **404** from the first sensor data **134a**. The set of user features **404** may indicate physical attributes of the user **102**, such as height, facial features, retinal features, pose estimation, etc. In this manner, the card device **120** may identify the user **102** for authentication and verification.

(86) Identifying an Intrusion of a Viewer

(87) In some cases, the card device **120** may be in a crowded location, such as in a public transport vehicle, or a crowded room, etc. In such cases, other people may be able to glance at the card device **120** and view the user information **138**. In some cases, the card device **120** may be accessed by a person other than the user **102**, such as an unauthorized person. In such cases, the other person

may be able to use the user information **138** in malicious actions, such as access personal information associated with the user **102** and/or transmit data from the card device **120** to other devices and impersonate the user **102**. Thus, it is desired to identify intruding viewers and in response, prevent them from accessing the user information **138** and using the card device **120** to perform data communications.

(88) In an example scenario, assume that the viewer **402** is detected by the sensors **126**. The card device **120** may receive the second sensor data **134b** from the sensors **126**. The second sensor data **134b** may indicate the presence of the viewer **402** around the card device **120**. The card device **120** may feed the sensor data **134b** to the machine learning algorithm **142**. The card device **120** (e.g., via the machine learning algorithm **142**) may extract a set of viewer features **408** from the sensor data **134b**. The viewer features **408** may indicate physical attributes of the viewer **402**, such as height, facial features, pose estimation, biometric features, etc.

(89) The card device **120** may compare each of the user features **404** with the counterpart viewer feature **408**. The card device **120** may determine whether each of the user features **404** corresponds to the counterpart viewer feature **408**. In certain embodiments, if the card device **120** determines that more than a threshold percentage of the user features **404** does not correspond to the counterpart viewer feature **408**, the card device **120** may determine that the viewer **402** is not the user **102**.

(90) In certain embodiments, the user features **404** may be represented by a user feature vector **410** comprising numerical values, and the viewer features **408** may be represented by a viewer feature vector **412** comprising numerical values. The card device **120** may determine a Euclidean distance between the user feature vector **410** and the viewer feature vector **412**. If the Euclidean distance is more than a threshold distance (e.g., more than 0.1, 0.01, etc.), the card device **120** may determine that the viewer **402** is not the user **102**.

(91) In certain embodiments, comparing each of the extracted user features **404** with the counterpart viewer feature **408** is in response to comparing each of the numerical values in the user feature vector **410** with a counterpart numerical value from among the numerical values of the viewer feature vector **412**.

(92) In certain embodiments, determining that the user **102** is not the viewer **402** may be in response to determining that more than the threshold percentage of the numerical values of the user feature vector **410** does not correspond to the counterpart numerical values from among the numerical values of the viewer feature vector **412**.

(93) The card device **120** may determine a distance **406** between the viewer **402** and the card device **120**. The card device **120** may compare the distance **406** to the threshold distance **140**. The threshold distance **140** may be maximum distance from which the user information **138** on the card device **120** is identifiable. The threshold distance **140** may be determined based on the font size of the user information **138**.

(94) If the card device **120** determines that the distance **406** is less than the threshold distance **140** and that the viewer **402** is not the user **102**, the card device **120** may dynamically mask the user information **138** displayed on the display field **128**, similar to that described in FIG. 2. In certain embodiments, the card device **120** may also disable data communicate for the card device **120**.

(95) In certain embodiments, the card device **120** may determine whether the user **102** or another person is carrying the card device **120** or is accessing the card device **120** based on pose estimation data (**220** and **224** in FIG. 2), similar to that described in FIG. 2. If the card device **120** determines that the other person **402** is accessing or carrying the card device **120**, the card device **120** may dynamically mask the user information **138** on the display field **128** and/or disable data communicate for the card device **120**.

(96) Method for Altering Card Device Attributes in Response to Detecting a Proximity Intrusion

(97) FIG. 5 illustrates an example flowchart of a method **500** for altering card device **120** attributes in response to a proximity intrusion of a viewer according to certain embodiments of the present

disclosure. Modifications, additions, or omissions may be made to method **500**. Method **500** may include more, fewer, or other operations. For example, operations may be performed in parallel or in any suitable order. While at times discussed as the system **100**, card device **120**, or components of any of thereof performing operations, any suitable system or components of the system may perform one or more operations of the method **300**. For example, one or more operations of method **300** may be implemented, at least in part, in the form of software instructions **132**, **158** of FIG. **1**, stored on non-transitory, tangible, machine-readable media (e.g., memory **130**, **156** of FIG. **1**) that when run by one or more processors (e.g., processor **122**, **152** of FIG. **1**) may cause the one or more processors to perform operations **502-520**.

(98) At operation **502**, the card device **120** receives a first sensor data **134a** from at least one sensor **126**, where the first sensor data **134a** indicates presence of a user **102**. For example, the sensors **126** may capture sensor data **134a** continuously, periodically, or on demand, or in response to a trigger event when the sensors **126** detect an object within their detection range.

(99) At operation **504**, the card device **120** extracts a set of user features **404** from the first sensor data **134a**, where the set of user features **404** indicates physical attributes of the user **102**. For example, the card device **120** may feed the sensor data **134a** to the machine learning algorithm **142** to extract the set of user features **404**.

(100) At operation **506**, the card device **120** receives a second sensor data **134b** from the at least one sensor **126**, where the second sensor data **134b** indicates presence of a viewer **402**. For example, the sensors **126** may capture the sensor data **134b** continuously, periodically, on demand, or in response to a trigger event when the sensors **126** detect an object within their detection range.

(101) At operation **508**, the card device **120** extracts a set of viewer features **408** from the second sensor data **134b**, where the set of viewer features **408** indicates physical attributes of the viewer **402**. For example, the card device **120** may feed the sensor data **134b** to the machine learning algorithm **142** to extract the set of viewer features **408**.

(102) At operation **510**, the card device **120** compares the set of user features **404** with the set of viewer features **408**. For example, the card device **120** (e.g., via the machine learning algorithm **142**) may perform a vector comparison operation between the user feature vector **410** that represents the user features **404** and the viewer feature vector **412** that represents the viewer features **408**, similar to that described in FIG. **4**. In one example, the card device **120** may compare each user feature **404** with a counterpart viewer feature **408**. In the same or another example, the card device **120** may determine a Euclidean distance between the user feature vector **410** and the viewer feature vector **412**.

(103) At operation **512**, the card device **120** determines whether more than a threshold percentage of the user features **404** correspond to the counterpart viewer features **408**. In certain embodiments, the card device **120** may determine whether the Euclidean distance between the user feature vector **410** and the viewer feature vector **412** is less than a threshold distance, similar to that described in FIG. **4**. If the card device **120** determines that more than a threshold percentage of the user features **404** correspond to the counterpart viewer features **408**, method **500** proceeds to operation **514**.

Otherwise, method **500** proceeds to operation **516**.

(104) At operation **514**, the card device **120** may allow data communications for the card device **120**. For example, the card device **120** may allow the card device **120** to communicate data with other devices. The card device **120** may also not mask or unmask the user information **138** displayed on the card device **120**. In response, if the user **102** wants to use the card device **120** to interact with another device, the card device **120** may communicate data (e.g., user information **138**) to the other device and generate the interaction session data **146**. The card device **120** may communicate the interaction session data **146** to the backend server **150** and/or to the blockchain network **180** to be distributed among the notes **182a-n**.

(105) At operation **516**, the card device **120** determines that the user **102** is not the viewer **402**. At operation **518**, the card device **120** determines the distance **406** between the viewer **402** and the

card device **120**. For example, the card device **120** may use a distance measuring feature in a camera sensor **126** and/or a light detection and ranging sensor **126** to detect the distance **406** between the viewer **402** and the card device **120**.

(106) At operation **520**, the card device **120** determines whether the distance **406** is less than a threshold distance **140**. If it is determined that the distance **406** is less than the threshold distance **140**, method **500** proceeds to operation **522**. Otherwise, method **500** returns to operation **506** to evaluate other people/viewers **402** around the card device **120**.

(107) At operation **522**, the card device **120** dynamically masks the user information **138** displayed on the card device **120**, similar to that described in FIGS. **1-4**. At operation **524**, the card device **120** disables the data communications for the card device **120**. In response, the card device **120** may not be able to interact with other devices.

(108) While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated with another system or certain features may be omitted, or not implemented.

(109) In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

(110) To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 112(f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. A system for dynamically changing attributes of a card device, comprising: at least one sensor configured to capture sensor data, wherein the sensor data provides information about an environment around a card device, wherein the information about the environment indicates objects located within the environment around the card device; a display field configured to display information, comprising a name or a number associated with a first user; a processor operably coupled to the at least one sensor and the display field, and configured to: receive a first sensor data from the at least one sensor, wherein the first sensor data indicates presence of the first user around the card device; extract a set of user features from the first sensor data, wherein the set of user features indicate physical attributes of the first user; receive a second sensor data from the at least one sensor, wherein the second sensor data indicates presence of a viewer around the card device; extract a set of viewer features from the second sensor data, wherein the set of viewer features indicate physical attributes of the viewer; compare at least one of the set of user features with a counterpart feature from among the set of viewer features; determine that the more than a threshold percentage of the set of user features does not correspond to counterpart features from among the set of viewer features; determine that the viewer is not the first user in response to determining that more than the threshold percentage of the set of user features does not correspond to the counterpart features from among the set of viewer features; determine a distance between the viewer and the card device; compare the determined distance with a threshold distance; determine

- that the determined distance is less than the threshold distance; and in response to determining that the determined distance is less than the threshold distance and that the viewer is not the first user, dynamically mask the information displayed on the display field.
2. The system of claim 1, wherein the processor is further configured to, in response to determining that the determined distance is less than the threshold distance and that the viewer is not the first user, disable the card device from being used for data communication.
 3. The system of claim 1, wherein the processor is further configured to: determine a first pose estimation data associated with the first user based at least in part upon the first sensor data, wherein the first sensor data comprises an image feed of the first user; receive a fourth sensor data from the at least one sensor, wherein the fourth sensor data comprises information about a second user, wherein the information about the second user comprises a second pose estimation associated with the second user; detect a second pose estimation data associated with the second user based at least in part upon the fourth sensor data; compare the first pose estimation data with the second pose estimation data; determine that the first pose estimation data does not correspond to the second pose estimation data; and in response to determining that the first pose estimation data does not correspond to the second pose estimation data, dynamically mask the information displayed on the display field.
 4. The system of claim 1, wherein the at least one sensor comprises a camera, a sound sensor, a light sensor, a humidity sensor, a biometric sensor, or a pressure sensor.
 5. The system of claim 1, wherein: the set of user features is represented by a user vector comprising a first set of numerical values; the set of viewer features is represented by a viewer vector comprising a second set of numerical values; and comparing the at least one of the extracted user features with the counterpart feature from among the extracted viewer features is in response to comparing at least one of the first set of numerical values with a counterpart numerical value from among the second set of numerical values.
 6. The system of claim 5, wherein determining that the first user is not the viewer is in response to determining that more than the threshold percentage of the first set of numerical values does not correspond to the counterpart numerical value from among the second set of numerical values.
 7. The system of claim 1, wherein the processor is further configured to: detect that a message is received at the card device; and dynamically mask the message in response to determining that the determined distance is less than the threshold distance and that the viewer is not the first user.
 8. A method for dynamically changing attributes of a card device, comprising: receiving a first sensor data from at least one sensor associated with a card device, wherein the first sensor data indicates presence of the first user around the card device; extracting a set of user features from the first sensor data, wherein the set of user features indicate physical attributes of the first user; receiving a second sensor data from the at least one sensor, wherein the second sensor data indicates presence of a viewer around the card device; extracting a set of viewer features from the second sensor data, wherein the set of viewer features indicate physical attributes of the viewer; comparing at least one of the set of user features with a counterpart feature from among the set of viewer features; determining that the more than a threshold percentage of the set of user features does not correspond to counterpart features from among the set of viewer features; determining that the viewer is not the first user in response to determining that more than the threshold percentage of the set of user features does not correspond to the counterpart features from among the set of viewer features; determining a distance between the viewer and the card device; comparing the determined distance with a threshold distance; determining that the determined distance is less than the threshold distance; and in response to determining that the determined distance is less than the threshold distance and that the viewer is not the first user, dynamically masking the information displayed on a display field.
 9. The method of claim 8, further comprising, in response to determining that the determined distance is less than the threshold distance and that the viewer is not the first user, disabling the

card device from being used for data communication.

10. The method of claim 8, further comprising: determining a first pose estimation data associated with the first user based at least in part upon the first sensor data, wherein the first sensor data comprises an image feed of the first user; receiving a fourth sensor data from the at least one sensor, wherein the fourth sensor data comprises information about a second user, wherein the information about the second user comprises a second pose estimation associated with the second user; detecting a second pose estimation data associated with the second user based at least in part upon the fourth sensor data; comparing the first pose estimation data with the second pose estimation data; determining that the first pose estimation data does not correspond to the second pose estimation data; and in response to determining that the first pose estimation data does not correspond to the second pose estimation data, dynamically masking the information displayed on the display field.

11. The method of claim 8, wherein the at least one sensor comprises a camera, a sound sensor, a light sensor, a humidity sensor, a biometric sensor, or a pressure sensor.

12. The method of claim 8, wherein: the set of user features is represented by a user vector comprising a first set of numerical values; the set of viewer features is represented by a viewer vector comprising a second set of numerical values; and comparing the at least one of the extracted user features with the counterpart feature from among the extracted viewer features is in response to comparing at least one of the first set of numerical values with a counterpart numerical value from among the second set of numerical values.

13. The method of claim 12, wherein determining that the first user is not the viewer is in response to determining that more than the threshold percentage of the first set of numerical values does not correspond to the counterpart numerical value from among the second set of numerical values.

14. The method of claim 8, further comprising: detecting that a message is received at the card device; and dynamically masking the message in response to determining that the determined distance is less than the threshold distance and that the viewer is not the first user.

15. A non-transitory computer-readable medium storing instructions that when executed by a processor, cause the processor to: receive a first sensor data from at least one sensor associated with a card device, wherein the first sensor data indicates presence of the first user around the card device; extract a set of user features from the first sensor data, wherein the set of user features indicate physical attributes of the first user; receive a second sensor data from the at least one sensor, wherein the second sensor data indicates presence of a viewer around the card device; extract a set of viewer features from the second sensor data, wherein the set of viewer features indicate physical attributes of the viewer; compare at least one of the set of user features with a counterpart feature from among the set of viewer features; determine that the more than a threshold percentage of the set of user features does not correspond to counterpart features from among the set of viewer features; determine that the viewer is not the first user in response to determining that more than the threshold percentage of the set of user features does not correspond to the counterpart features from among the set of viewer features; determine a distance between the viewer and the card device; compare the determined distance with a threshold distance; determine that the determined distance is less than the threshold distance; and in response to determining that the determined distance is less than the threshold distance and that the viewer is not the first user, dynamically mask the information displayed on a display field.

16. The non-transitory computer-readable medium of claim 15, wherein the instructions further cause the processor to, in response to determining that the determined distance is less than the threshold distance and that the viewer is not the first user, disable the card device from being used for data communication.

17. The non-transitory computer-readable medium of claim 15, wherein the instructions further cause the processor to: determine a first pose estimation data associated with the first user based at least in part upon the first sensor data, wherein the first sensor data comprises an image feed of the

first user; receive a fourth sensor data from the at least one sensor, wherein the fourth sensor data comprises information about a second user, wherein the information about the second user comprises a second pose estimation associated with the second user; detect a second pose estimation data associated with the second user based at least in part upon the fourth sensor data; compare the first pose estimation data with the second pose estimation data; determine that the first pose estimation data does not correspond to the second pose estimation data; and in response to determining that the first pose estimation data does not correspond to the second pose estimation data, dynamically mask the information displayed on the display field.

18. The non-transitory computer-readable medium of claim 17, wherein: the first pose estimation data is represented by a first pose estimation vector comprising a third set of numerical values; the second pose estimation data is represented by a second pose estimation vector comprising a fourth set of numerical values; determining that the first pose estimation data does not correspond to the second pose estimation data is in response to: comparing the first pose estimation vector with the second pose estimation vector; and determining that more than a threshold numerical values of the third set of numerical values does not correspond to counterpart numerical values from among the fourth set of numerical values.

19. The non-transitory computer-readable medium of claim 15, wherein the distance between the viewer and the card device is determined from an image of the viewer captured by a camera sensor from among the at least one sensor.

20. The non-transitory computer-readable medium of claim 15, wherein: the set of user features is represented by a user vector comprising a first set of numerical values; the set of viewer features is represented by a viewer vector comprising a second set of numerical values; and comparing the at least one of the extracted user features with the counterpart feature from among the extracted viewer features is in response to comparing at least one of the first set of numerical values with a counterpart numerical value from among the second set of numerical values.
